

Emmanuel Kuh

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PROFESSIONAL SUMMARY

Applied quantitative economist and data scientist specializing in causal inference, machine learning, and statistical modeling. Experienced delivering data-driven consulting, building reproducible analytical pipelines in R, Python, SQL, and Stata, and translating complex data into actionable insights across health, finance, technology, and public policy.

EDUCATION

Towson University

M.S. Econometrics and Quantitative Economics (Economic Analytics).

Towson, MD

Expected December 2026

Relevant Coursework: Causal Inference, Machine Learning for Economics, Econometrics, Computational Economics, Impact Evaluation, Mathematical Economics

University of Maryland, College Park

M.P.S. Public Administration. GPA: 3.88/4.00.

College Park, MD

2024 – 2025

University of Education, Winneba

B.A. Political Science Education and Economics.

Winneba, Ghana

July 2020

RESEARCH INTERESTS

- Causal Machine Learning, Causal Inference, Health Data Science, Statistical Machine Learning, Computational Social Science, Applied Econometrics, Data-Driven Policy Evaluation

RESEARCH PRESENTATIONS

- Medicare and Mental Health: Causal Evidence from a Fuzzy RD Design, Towson University (2026)
- Socioeconomic Determinants of Homicide in Baltimore Neighborhoods, Towson University
- Causal Machine Learning and Health Insurance: Kreif et al. (2020)
- Regression Discontinuity and the Minimum Drinking Age: Carpenter and Dobkin (2011)
- Deworming and Education Externalities: Miguel and Kremer (2004)

PROJECTS

Medicare and Mental Health: Causal Evidence from a Fuzzy Regression Discontinuity Design

2026 – Present

Working Paper | Supervisor: Professor Juergen Jung | Towson University

- Exploit age-65 Medicare eligibility threshold using RAND HRS waves 9–14 (2008–2018): 32,894 person-wave observations from 12,709 respondents; 6,352 couple-wave observations for spillover analysis.
- Implement local-linear fuzzy RD/IV with triangular-kernel weighting and clustered errors; 63.5 pp first-stage increase in coverage.
- Validate via alternative bandwidths, placebo cutoffs, density tests, covariate-continuity, and donut-hole models.

Labor Supply Effects of Spousal Long-Term Care Shocks

Independent Ongoing Research Project | Towson University

- Apply Sun-Abraham (2021) staggered DiD to 80,423 HRS observations (2004–2018): 1,945 treated vs. 18,599 never-treated; exploit first spousal transition to 2+ ADL limitations.
- Examine employment, hours, paid-care use, and caregiving outcomes using individual and wave fixed effects.

Weather Intelligence Platform

2026

Independent Project | JavaScript, HTML, CSS, Open Meteo, NOAA, OpenStreetMap APIs

- Design and develop a 2,800+ line full-stack weather intelligence platform integrating 3 live APIs with interactive maps, analytics dashboards, voice alerts, and responsive visualizations for worldwide weather monitoring.
- Engineer 30+ production features including global location search, support for all 50 U.S. states and 500+ cities, real-time analytics, city comparison, geolocation, and offline browser storage.

Socioeconomic Determinants of Homicide in Baltimore Neighborhoods

Sole-Authored Paper | Towson University

- Analyze 54 Baltimore CSAs using BNIA-JFI data; implement OLS and stepwise AIC selection; identify unemployment and juvenile arrest rates as strongest predictors (R-squared 0.669, $p < 0.001$).

Credit Default Prediction Using Machine Learning github.com/kuhemmanuel9-sudo

Independent Project | Python, scikit-learn

- Build end-to-end pipeline comparing logistic regression and neural networks; evaluate via ROC-AUC, precision, recall, and F1.

EXPERIENCE

Economic Researcher and Graduate Assistant

Towson University, Department of Economics

2025 – Present

Towson, MD

- Conduct faculty-supervised and independent empirical research applying causal identification strategies to large longitudinal health datasets.
- Build reproducible research pipelines in R, Python, Stata, and SQL, producing econometric estimates, diagnostic analyses, visualizations, and manuscript exhibits.
- Advance independent research using Sun-Abraham (2021) staggered DiD on spousal LTC shocks across 80,423 HRS person-wave observations.

Graduate Assistant

Aug 2025 – Present
 Towson University, M.S. Economic Analytics Program
 Towson, MD

- Design and execute global recruitment analytics campaigns expanding prospective student reach across 30+ countries.
- Analyze student feedback and translate findings into evidence-based recommendations for program leadership.

Thurgood Marshall Research Fellow

May 2025 – Aug 2025
 University of Maryland School of Public Policy
 College Park, MD

- Led policy research on racial and social equity, synthesizing evidence to develop recommendations for public leadership.

HR and Program Development Assistant

May 2025 – Jun 2025
 University of Maryland Facilities Management
 College Park, MD

- Analyzed workforce data supporting HR reporting, compliance, and organizational planning across an 800+ employee operation.

Professional Teacher

2019 – 2024
 Ghana Education Service
 Ghana

- Taught Economics and Government to 14,000+ students, achieving 95% WASSCE pass rates; mentored 500+ students for university admission.

SKILLS

Econometrics and Causal Inference: Regression Discontinuity (Fuzzy RD), Instrumental Variables (IV/2SLS), Difference-in-Differences (DiD), Sun-Abraham Staggered DiD, Event Studies, Fixed Effects, Panel Data, Survey-Weighted Estimation, Treatment Effect Heterogeneity

Machine Learning and Statistics: Causal Forests, Generalized Random Forests, Gradient Boosting, Random Forests, Neural Networks, Logistic Regression, Principal Component Analysis (PCA), Cross-Validation

Programming and Data: R, Python, SQL, Stata, JavaScript

Data Visualization and Tools: Tableau, Power BI, Excel, Git, Jupyter Notebook, RStudio

AWARDS & HONORS

- Summer Research Grant, Economics, Towson University (2026)
- Thurgood Marshall Research Fellowship, University of Maryland School of Public Policy (2025)
- Distinguished Public Administration Graduate Student, University of Maryland (2025)
- John J. Sexton Fellowship Award, University of Maryland (2025)
- Dean's Fellowship Award, University of Maryland (2024)
- Willard Hackerman Fellowship Award, University of Maryland (2024)

ACTIVITIES & LEADERSHIP

Towson University, College of Business and Economics

Aug 2025 – Present
 Vice President, Graduate Student Association

- Review and allocate research, travel, and professional development funding for 3,000+ graduate students; serve on Graduate Curriculum Committee, Academic Assessment Council, and Towson University International Committee.

University of Maryland, College Park

May 2025 – Aug 2025
 Graduate Senator

- Advocated for 10,000+ graduate students; applied statistical analysis to identify funding equity gaps and delivered evidence-based recommendations to university leadership.

U.S. Department of State, U.S. Department of Defense, U.S. House of Representatives, and Maryland General Assembly

Oct 2024 – May 2025
 Government and Institutional Engagement

- Engaged senior diplomats, defense officials, and legislators on foreign relations, national security, and priority bills across housing, health, education, and climate policy.

CERTIFICATIONS

- Certificate in AI and Career Empowerment, Robert H. Smith School of Business, University of Maryland (2025)
- Teaching License, National Teaching Council, Ghana (2022)
- Social and Behavioral Research – Basic/Refresher, CITI Program (2025)